

Economic Growth and Inclusive Development

Rounak Raj
Be Cse
Chitkara University
Rajpura, Punjab
rounak2187.be22@chitkara.edu.in

Jatin Kansal
Be Cse
Chitkara University
Rajpura, Punjab
jatin1710.be22@chitkara.edu.in

Rohan Jain
Be Cse
Chitkara University
Rajpura, Punjab
rohan2175.be22@chitkara.edu.in

Rahul Thakur
Be Cse
Chitkara University
Rajpura, Punjab
rahul704.be22@chitkara.edu.in

Abstract

This study investigates the evolving nature of economic growth and underscores the growing necessity for a new development model that prioritizes inclusivity, employment generation, and environmental sustainability. Traditional theories of economic development largely viewed industrialization as the central engine of growth, often relegating agriculture to a passive or transitional role. In these frameworks, agriculture was primarily expected to supply surplus labor, raw materials, and savings to the industrial sector, with limited attention paid to its own productivity, resilience, or long-term contribution to development. However, contemporary global and national experiences increasingly challenge this linear approach, particularly in the context of developing economies. In recent decades, two critical challenges—persistent unemployment and mounting environmental degradation—have emerged as major constraints to conventional growth strategies. Rapid technological change, automation, and capital-intensive industrialization have weakened the capacity of the industrial sector to absorb surplus labor, especially in countries with large populations dependent on agriculture. At the same time, unbalanced growth patterns have intensified ecological stress, manifested through land degradation, water scarcity, and climate vulnerability. These structural shifts demand a fundamental rethinking of growth models that move beyond purely industrial-led trajectories and instead emphasize sustainability and broad-based participation. Against this backdrop, the paper critically examines the changing role of agriculture in the process of economic transformation, with particular reference to developing economies such as India. It explores how advancements in agricultural technology, shifts in labor productivity, and changing rural–urban dynamics have altered the traditional pathways of structural transformation. The study also highlights the growing interdependence between agriculture, industry, and services, noting that weaknesses in one sector often constrain the growth potential of others. Special attention is given to emerging sectoral imbalances, where the expansion of the service sector has not been accompanied by adequate employment creation or a strong industrial base. Further, the study undertakes a comparative analysis of growth patterns across Indian states to examine how different sectoral compositions influence development outcomes. By analyzing variations in agricultural performance, industrial expansion, and service-sector growth, the paper seeks to understand the conditions under which economic growth becomes more inclusive and socially balanced. The findings suggest that states which have successfully strengthened agriculture while gradually fostering industrial development tend to achieve better employment outcomes, reduced income disparities, and greater economic resilience. The study concludes that agriculture-led growth, when strategically supported and subsequently complemented by industrial development, offers the most viable and inclusive pathway for long-term economic transformation. Rather than viewing agriculture as a residual sector, the paper argues for its central role in generating employment, ensuring food security, stabilizing rural livelihoods, and supporting sustainable use of natural resources. Ultimately, the paper calls for a new growth paradigm that integrates agriculture with industry and services, promotes decent employment opportunities, and aligns economic progress with

environmental sustainability. Such a holistic approach is essential for achieving inclusive and sustainable development in the contemporary economic landscape.

Keywords

Economic Growth; Inclusive Development; Agriculture; Employment; Sustainability; Structural Transformation; Industrialization; India

Introduction

Economic development has traditionally been conceptualized as a process of structural transformation in which resources—particularly labor and capital—shift from low-productivity agriculture to higher-productivity sectors such as industry and services. This transition was widely regarded as a necessary condition for accelerating economic growth, improving productivity, and raising income levels. Classical and neoclassical development theories emphasized industrial expansion as the primary engine of growth, with agriculture largely assigned a supportive or residual role. In this framework, agriculture was expected to provide food, raw materials, surplus labor, and savings to fuel industrialization, after which its relative importance would gradually decline.

However, the contemporary global economic landscape presents a set of challenges that call into question the adequacy of this traditional development trajectory. One of the most pressing concerns is employment generation. Advances in technology, automation, and the adoption of capital-intensive production processes have significantly reduced the capacity of industry to absorb labor, particularly in developing economies with large and growing workforces. As a result, economic growth has increasingly become decoupled from employment creation, leading to the phenomenon of jobless or low-employment growth. This has intensified income inequalities and limited the inclusiveness of growth.

Simultaneously, environmental sustainability has emerged as a critical constraint on long-term economic development. Issues such as climate change, water scarcity, land degradation, biodiversity loss, and excessive resource extraction have exposed the ecological costs of conventional growth models. Agriculture, which is highly dependent on natural resources, is particularly vulnerable to these environmental stresses. Yet, it also holds significant potential for promoting sustainable development through improved resource management, climate-resilient practices, and renewable bio-based activities.

In this changing context, the role of agriculture in the development process is being fundamentally re-evaluated. Rather than being viewed as a backward or declining sector, agriculture is increasingly recognized as a potential driver of economic growth, employment generation, and environmental sustainability. Improvements in agricultural productivity can stimulate demand for industrial goods and services, raise rural incomes, reduce poverty, and create forward and backward linkages with non-agricultural sectors. Moreover, agriculture remains a major source of livelihood for a large proportion of the population in developing countries, making it central to achieving inclusive growth.

This reorientation has important implications for countries like India, where structural transformation has followed an uneven path.

While the service sector has expanded rapidly, industrial growth and employment absorption have lagged behind, and agriculture continues to employ a substantial share of the workforce. These structural imbalances raise critical questions about the sustainability and inclusiveness of the existing growth model and highlight the need to explore alternative development pathways.

Against this backdrop, the present paper aims to examine whether a new growth model is required to address contemporary economic, social, and environmental challenges. It seeks to analyse the potential role of agriculture as a leading sector in promoting inclusive development, generating employment, and ensuring sustainable use of natural resources. By reassessing the conventional hierarchy between agriculture, industry, and services, the study contributes to the broader debate on redefining development strategies in the twenty-first century.

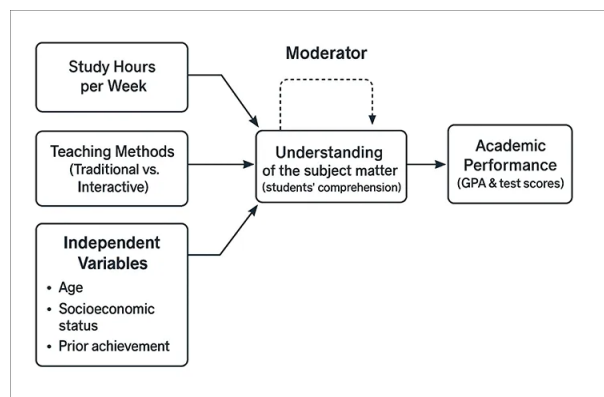


Fig. 1. Conceptual framework

Objective of the work

The primary objective of this study is to critically examine the changing nature of economic growth and assess the need for a new development model that ensures inclusive and sustainable outcomes. In pursuit of this broad aim, the study seeks to achieve the following specific objectives:

To examine the traditional model of economic development that emphasizes structural transformation from agriculture to industry and services, and to assess its relevance in the present global and national context.

To analyse the emerging challenges of contemporary growth patterns, particularly with respect to employment generation, automation, capital-intensive production, and environmental sustainability.

To re-evaluate the role of agriculture in economic development, moving beyond its conventional perception as a backward or residual sector, and assessing its potential contribution to growth, employment, and sustainability.

To study the interlinkages between agriculture, industry, and services in the process of economic transformation, with special reference to developing economies such as India.

To examine patterns of sectoral growth and structural change and their implications for inclusive development and income distribution.

To assess whether agriculture-led growth can provide a more balanced and inclusive growth pathway, particularly when complemented by industrial development.

To contribute to the formulation of a new growth paradigm that integrates agriculture with broader development strategies while ensuring sustainable use of natural resources.

Literature review

Early theories of economic development placed strong emphasis on the process of structural transformation, viewing development primarily as a shift of labour and resources from agriculture to industry and, eventually, to services. Classical development models conceptualized agriculture as a traditional, low-productivity sector whose main function was to support industrialization by supplying surplus labour, food, and capital. Development, in this view, was achieved when labour moved out of agriculture into modern industrial sectors characterized by higher productivity, better wages, and faster technological progress. This framework strongly

influenced development policies in many post-colonial and developing economies during the mid-twentieth century.

Subsequent research, however, began to question the assumption that agriculture plays only a passive or transitional role in economic growth. Studies during the 1960s and 1970s highlighted the importance of agriculture as an active contributor to development rather than merely a reservoir of surplus labour. Scholars emphasized the existence of strong forward and backward linkages between agriculture and other sectors. Agricultural growth was shown to stimulate demand for industrial inputs such as fertilizers, machinery, and irrigation equipment, while simultaneously supplying raw materials to agro-based industries. Additionally, increased agricultural income was found to boost rural consumption, thereby expanding markets for industrial and service-sector goods. This body of literature underscored the role of agriculture in reducing poverty, improving food security, and generating broad-based growth.

Later empirical studies further strengthened the case for agriculture-led development, particularly in countries where a large proportion of the population depended on rural livelihoods. It was argued that neglecting agriculture could lead to structural imbalances, regional disparities, and persistent rural poverty. Countries that experienced early industrial growth without sufficient agricultural development often faced food shortages, rising inequality, and limited employment absorption. As a result, a more balanced growth strategy that recognized agriculture as a productive and dynamic sector gained increasing attention in development discourse. In more recent decades, the focus of the literature has shifted toward emerging challenges associated with contemporary growth patterns. One major concern is the phenomenon of jobless or low-employment growth. Technological advancements, automation, and the increasing use of capital-intensive production techniques have enhanced productivity but significantly reduced the capacity of both industry and services to absorb labour. This trend has weakened the traditional expectation that labour released from agriculture would be smoothly absorbed by non-agricultural sectors. As a consequence, many developing economies continue to experience high levels of informal employment, underemployment, and income inequality.

At the same time, environmental sustainability has become a central concern in development literature. Rapid industrialization and resource-intensive growth models have contributed to environmental degradation, climate change, water scarcity, and land exhaustion. In this context, agriculture has acquired a renewed significance—not only as a source of employment and income but also as a sector closely linked to ecological stability. Recent studies highlight agriculture's expanding role in sustainable development through climate-resilient farming practices, renewable energy generation (such as biofuels), carbon sequestration, and sustainable land and water management.

Contemporary literature therefore increasingly advocates for an integrated growth model that recognizes the interdependence between agriculture, industry, and services. Rather than following a linear development path, scholars argue for a coordinated approach in which agricultural development supports industrial growth, employment generation, and environmental sustainability. This perspective is particularly relevant for developing countries like India, where structural transformation has been uneven and agriculture continues to employ a substantial share of the workforce.

Building on these theoretical and empirical contributions, the present study integrates traditional and modern perspectives on economic growth to evaluate the need for a new development model. By reassessing the role of agriculture in the context of employment challenges and environmental constraints, this study seeks to contribute to the evolving debate on inclusive and sustainable economic development.

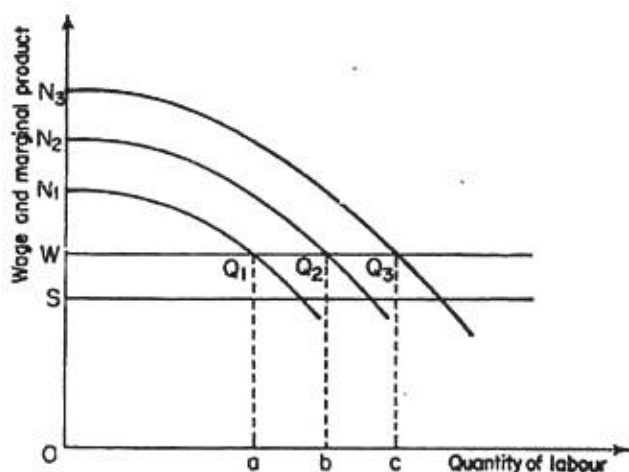


Fig. 14 : The Lewis Model

Fig. 2. The Lewis Model

Limitations of Existing Work

Despite the extensive body of literature on economic growth and structural transformation, several limitations can be identified in existing studies, which justify the need for further research. First, a large part of the traditional development literature adopts a sectorally linear approach, assuming a sequential transition from agriculture to industry and then to services. This framework tends to oversimplify the development process and fails to account for the complex interdependencies among sectors. In many developing economies, including India, agriculture continues to employ a significant proportion of the workforce even as industry and services expand. Existing models often overlook this persistence and thereby provide an incomplete explanation of contemporary growth patterns. Second, much of the earlier work treats agriculture as a passive or residual sector, focusing primarily on its role in supplying surplus labour and resources to industry. This perspective neglects agriculture's potential as an independent driver of growth, employment, and demand generation. Studies that emphasize industrialization frequently understate the importance of improving agricultural productivity and rural incomes as prerequisites for inclusive development. Third, existing research has often placed greater emphasis on aggregate economic growth while paying limited attention to the quality of growth, particularly in terms of employment generation and income distribution. With the increasing prevalence of jobless growth driven by automation and capital-intensive production methods, many studies fail to adequately address the disconnect between growth and employment. As a result, the implications of sectoral growth patterns for labour absorption and livelihood security remain underexplored. Fourth, the environmental dimension of development has received insufficient attention in earlier growth models. Traditional studies largely ignored ecological constraints, assuming infinite availability of natural resources. Although recent literature has begun to incorporate sustainability concerns, there is still limited integration of environmental factors with sectoral growth strategies, particularly with respect to agriculture's dual role as both a resource-dependent and sustainability-enhancing sector. Fifth, many empirical studies rely on national-level analyses, which tend to mask significant regional and state-level variations. In a country like India, where states exhibit diverse growth experiences and sectoral compositions, such aggregate analyses limit the understanding of how different development pathways influence inclusiveness and sustainability. Comparative state-level assessments remain relatively scarce in the literature. Finally, existing work often examines agriculture, industry, and services in isolation, rather than within a unified and integrated analytical framework. This segmented approach restricts the ability to evaluate alternative growth models that promote balanced development through coordinated sectoral advancement. In view of these limitations, there is a clear need for research that re-evaluates conventional growth models by integrating agriculture more centrally into the development framework, accounting for employment and environmental challenges, and adopting a more holistic and inclusive perspective.

Research Gap

The review of existing literature reveals that while economic growth and structural transformation have been extensively studied, several critical gaps remain unaddressed. Traditional development theories have largely focused on a linear transition from agriculture to industry and services, offering limited insight into alternative growth pathways that may be more relevant under present economic and environmental conditions. As a result, much of the literature fails to adequately explain the contemporary development experience of many developing economies. First, there is a lack of comprehensive analysis that integrates agriculture as a central component of modern growth models. Although some studies acknowledge the importance of agriculture, they often continue to treat it as secondary to industrial and service-sector growth. There is insufficient research that systematically examines agriculture-led growth as a viable and sustained strategy for inclusive development. Second, existing studies have not sufficiently addressed the disconnect between economic growth and employment generation. While recent literature discusses jobless growth, few studies explicitly analyse how different sectoral growth patterns—particularly agriculture versus industry- or service-led growth—affect employment outcomes. This gap limits understanding of how growth strategies can be aligned with labour absorption and livelihood security. Third, the environmental dimension of growth remains inadequately integrated into analyses of structural transformation. Although sustainability has emerged as a key concern, limited research has examined how agriculture can simultaneously contribute to economic growth and environmental preservation. The dual role of agriculture as both a resource-intensive sector and a potential driver of sustainability is not sufficiently explored within growth models. Fourth, there is a scarcity of region- or state-level studies, especially in the Indian context, that compare sectoral growth patterns and their implications for inclusiveness and sustainability. National-level aggregates often conceal regional disparities, thereby restricting the evaluation of diverse development trajectories across states with varying agricultural, industrial, and service-sector compositions. Finally, much of the existing work analyses agriculture, industry, and services in isolation rather than within a holistic and interlinked framework. The absence of integrated approaches limits the ability to assess coordinated sectoral development and the sequencing of growth strategies necessary for balanced and inclusive outcomes. In light of these gaps, the present study seeks to contribute to the literature by reassessing the role of agriculture within a unified growth framework that addresses employment generation, sectoral balance, and environmental sustainability. By bridging traditional and contemporary perspectives, the study aims to offer new insights into the formulation of a more inclusive and sustainable growth model.

Research Objectives

The overall objective of this study is to examine the changing nature of economic growth and to assess the relevance of a new growth model that promotes inclusive and sustainable development. In order to achieve this broad aim, the study pursues the following specific objectives:

To examine the traditional growth model based on structural transformation, particularly the shift of resources from agriculture to industry and services, and to assess its limitations in the contemporary economic context.

To analyse emerging challenges associated with current growth patterns, with special reference to employment generation, automation, capital-intensive production, and environmental sustainability.

To re-evaluate the role of agriculture in economic development, highlighting its potential contribution to growth, employment creation, and inclusive development rather than viewing it merely as a residual sector.

To study the interrelationships between agriculture, industry, and services, and their combined role in shaping balanced and inclusive economic growth.

To examine sectoral growth patterns and structural changes, particularly in the context of developing economies such as India. To assess the potential of agriculture-led growth as a viable development strategy, especially when followed and supported by industrial development.

To explore the need for a new growth paradigm that integrates agriculture within a broader development framework while ensuring

environmental sustainability and efficient use of natural resources.

Changing Role of Agriculture in Development

In early development theories, agriculture was largely perceived as a low-productivity sector characterized by traditional methods of production and the presence of surplus labour. It was assumed that agriculture's primary function was to support industrial growth by releasing excess labour, supplying cheap food, and generating savings for capital accumulation. As economies developed, the share of agriculture in employment and output was expected to decline steadily, making way for industry and services to take the lead. This perception led to the relative neglect of agriculture in development policies, particularly in the initial stages of industrialization.

However, over time, this narrow view of agriculture has undergone significant transformation. Technological advancements have played a crucial role in redefining agriculture's contribution to economic development. The introduction of high-yielding varieties of seeds, expansion of irrigation facilities, mechanization, use of fertilizers and modern farming techniques have substantially improved agricultural productivity. These changes have demonstrated that agriculture is not inherently backward, but rather a dynamic sector capable of generating growth when supported by appropriate technology, infrastructure, and institutional frameworks. Modern agriculture has also emerged as a strong source of employment, especially in developing economies where a large portion of the population continues to depend on rural livelihoods. Beyond on-farm employment, agriculture generates opportunities in allied activities such as food processing, storage, transportation, marketing, and agro-based industries. These activities enhance value addition and create productive employment across the rural economy, thereby reducing poverty and income disparities. Another important aspect of the changing role of agriculture is its growing interconnection with other sectors of the economy. Agriculture establishes strong backward linkages by creating demand for industrial inputs such as fertilizers, pesticides, agricultural machinery, irrigation equipment, and energy. Simultaneously, it forms forward linkages by supplying raw materials to agro-processing industries, textiles, sugar, edible oils, and bio-based manufacturing. These linkages stimulate industrial growth and help integrate agriculture into the broader economic system rather than isolating it as a subsistence activity. Rising agricultural productivity and incomes also strengthen the consumption linkage. Increased rural incomes raise demand for manufactured goods and services such as consumer durables, healthcare, education, transportation, and financial services. This expansion of rural demand contributes to overall economic growth and provides a stable domestic market for industrial and service-sector expansion. In economies where domestic demand plays a critical role, agricultural growth becomes an important stabilizing force.

Furthermore, agriculture has acquired renewed importance in the context of sustainable development. With growing concerns over climate change, resource depletion, and environmental degradation, agriculture is being recognized not only as a sector affected by environmental challenges but also as one that can contribute to sustainable solutions. Practices such as climate-resilient farming, organic agriculture, water-efficient irrigation, renewable energy use, and sustainable land management highlight agriculture's potential role in environmental conservation and long-term economic resilience.

These multidimensional contributions underline the fact that agriculture can act as an effective growth engine, particularly in the early stages of development. Countries that prioritize agricultural development often experience more inclusive growth, as benefits are distributed across wider sections of society. Rather than being viewed as a transitional or declining sector, agriculture is increasingly regarded as a foundational pillar that supports industrialization, employment generation, food security, and sustainability.

In this evolving understanding, the role of agriculture is no longer confined to feeding the population but extends to driving economic transformation, promoting equity, and ensuring sustainable use of natural resources. This shift in perspective forms the basis for rethinking conventional growth models and designing development strategies that place agriculture at the center of inclusive and sustainable economic development.

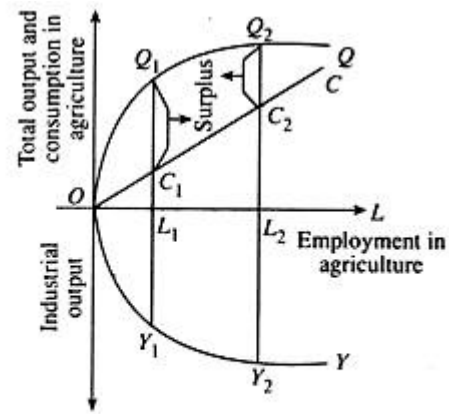


Fig. 1: Agriculture-Industry interdependence

Fig. 3. Role of Agriculture in Economic Development in India

Employment Challenge in Modern Growth

One of the most critical challenges confronting modern economic growth is the declining capacity of growth to generate adequate employment. While many economies have experienced rapid increases in output and productivity, this growth has not been accompanied by a proportionate rise in employment opportunities. This growing disconnect between economic growth and job creation has raised serious concerns about the inclusiveness and sustainability of contemporary development models. A major factor contributing to this employment challenge is the increasing reliance on capital-intensive and technology-driven production processes. Advances in automation, artificial intelligence, and digitization have significantly enhanced productivity in industry and services, but they have also reduced the demand for labour. Modern manufacturing relies more on machines than on human labour, limiting the absorption of workers displaced from traditional sectors such as agriculture. As a result, industrial growth no longer guarantees large-scale employment as it once did during earlier phases of development. The service sector, which has emerged as a dominant contributor to GDP in many developing economies, also presents mixed employment outcomes. While certain service industries such as information technology, finance, and professional services generate high incomes, they tend to be skill-intensive and employ a relatively small segment of the workforce. A significant proportion of the labour force lacks the education and skills required to access these jobs, leading to exclusion and widening income inequality. Thus, service-led growth often fails to provide sufficient employment opportunities for low- and semi-skilled workers. This situation has given rise to the phenomenon commonly described as jobless growth, where economic expansion occurs without commensurate employment creation. In developing countries, this problem is particularly severe due to the large influx of labour from rural areas and the continued dependence of a significant segment of the population on traditional livelihoods. When non-agricultural sectors are unable to absorb surplus labour, the result is persistent informal employment, underemployment, and precarious working conditions. Another dimension of the employment challenge relates to structural imbalance in growth patterns. In many economies, structural transformation has been incomplete, with labour moving out of agriculture at a faster pace than employment opportunities are created in industry and services. This has intensified urban unemployment, stress on urban infrastructure, and the expansion of informal sectors. The absence of coordinated sectoral development weakens the employment potential of overall growth. Furthermore, globalization and increased competition have also influenced employment outcomes. Firms facing competitive pressures often adopt labour-saving technologies to reduce costs, further limiting job creation. At the same time, labour market flexibility and informalization have increased, weakening employment security and social protection for workers. In this context, the role of agriculture assumes

renewed importance. Agriculture remains one of the largest employers in developing economies and has a greater capacity to generate livelihoods, particularly when linked with allied activities such as agro-processing, storage, transportation, and rural services. Strengthening agriculture can help absorb labour productively, stabilize rural incomes, and reduce distress migration. Unlike capital-intensive sectors, agriculture-led growth tends to be more employment-intensive and inclusive, especially in the early stages of development. The employment challenge in modern growth thus highlights the limitations of conventional development strategies that rely heavily on industrialization and service-sector expansion alone. It underscores the need for a new growth paradigm that places employment generation at the center of development planning, integrates agriculture with industry and services, and promotes balanced sectoral growth. Addressing employment challenges is not only essential for reducing poverty and inequality but also for ensuring social stability and sustainable long-term development.

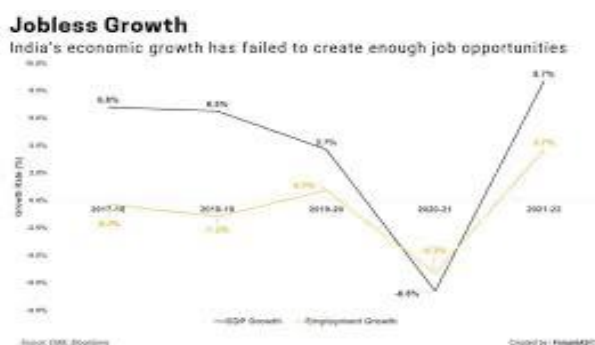


Fig. 4. Jobless Growth

Patterns of Growth in India

India's economic growth trajectory has evolved through distinct phases, reflecting changes in development priorities, policy orientation, and sectoral contributions. Unlike the linear paths suggested by classical development theories, India's experience demonstrates a more complex and uneven pattern of structural transformation, shaped by historical, institutional, and regional factors.

The first major phase of growth was driven primarily by agricultural development, particularly during the Green Revolution of the late 1960s and 1970s. The introduction of high-yielding variety seeds, expansion of irrigation facilities, increased use of fertilizers, and institutional support through minimum support prices significantly improved agricultural productivity. This phase not only enhanced food security but also raised rural incomes and generated surplus demand for industrial goods and services. Agricultural growth during this period played a crucial role in reducing poverty and laid the foundation for broader economic expansion, especially in states such as Punjab, Haryana, and western Uttar Pradesh.

The second phase of growth witnessed a gradual shift toward industrial development, especially after the economic reforms initiated in the early 1990s. Liberalization, privatization, and globalization policies aimed to promote industrial efficiency, attract investment, and integrate the Indian economy with global markets. Manufacturing and infrastructure development became central to growth strategies. However, industrial growth during this phase was increasingly capital-intensive and uneven across regions, limiting its capacity to absorb labour on a large scale. While some states benefited from industrial expansion, others lagged behind due to inadequate infrastructure, policy constraints, or weaker institutional capacity.

The third phase of growth, emerging more prominently in the post-reform period, has been characterized by the rapid expansion of the service sector, particularly information technology, finance, telecommunications, and business services. This services-led growth has significantly contributed to India's GDP and global integration. Nevertheless, service-sector growth has been largely skill-intensive, generating high incomes but relatively fewer

employment opportunities for the broader workforce. As a result, growth during this phase has been associated with rising income disparities and regional imbalances.

At the state level, India exhibits considerable diversity in growth patterns and development pathways. Some states achieved relatively inclusive growth by strengthening agriculture first, followed by gradual industrial development. These states benefited from higher rural incomes, better labour absorption, and stronger domestic demand. Others adopted growth strategies focused primarily on industry or services without a solid agricultural foundation. While such strategies resulted in higher per capita incomes, they often led to increased inequality, urban-centric development, and persistent rural distress.

States that relied heavily on service-led growth, particularly those centered around urban knowledge hubs, experienced rapid economic expansion but limited employment generation for low- and semi-skilled workers. In contrast, states with sustained agricultural performance and agro-based industrialization managed to create more balanced growth outcomes. These variations highlight the importance of sectoral composition and sequencing in shaping development outcomes.

The Indian experience thus underscores that the sequence of development matters significantly. Growth paths that prematurely shift away from agriculture or bypass industrialization tend to face challenges related to employment, inequality, and sustainability. In contrast, agriculture-led growth that strengthens rural economies, generates demand, and is subsequently complemented by industrial development appears to offer a more balanced and inclusive pathway. Overall, India's patterns of growth reveal the limitations of conventional development models that emphasize either industrial or service-led growth in isolation. The evidence suggests that a development strategy rooted in agricultural strengthening, followed by diversified industrial expansion and supported by services, is more likely to ensure inclusive,

employment-generating, and sustainable economic development. This insight has important implications for rethinking growth strategies in India and other developing economies facing similar structural challenges.

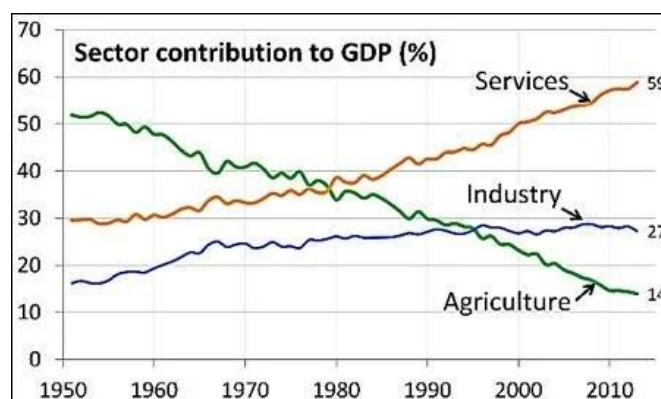


Fig 4. Phases of Economic Growth in India

Sustainability and Resource Use

Sustainability has emerged as a central concern in contemporary economic development, particularly in the context of increasing environmental stress and resource constraints. Economic growth strategies that focus solely on output expansion without considering ecological limits have led to significant degradation of natural resources. In this context, sustainability refers to the ability to meet present development needs without compromising the capacity of future generations to meet their own needs. Agriculture occupies a crucial position in this debate, as it is deeply dependent on natural resources such as land, water, air, and biodiversity. Agriculture is one of the most resource-intensive sectors of the economy. It consumes a substantial share of freshwater resources and relies heavily on soil quality and climatic conditions. In many regions, excessive extraction of groundwater, inefficient irrigation practices, and expansion of cultivation into fragile ecosystems have resulted in declining water tables, soil erosion, salinization, and loss of biodiversity. The overuse of chemical fertilizers and pesticides has further aggravated environmental problems by polluting soil and

water bodies, reducing soil fertility in the long run, and posing risks to human health. In addition to resource degradation, agricultural activities contribute to environmental challenges such as climate change. Practices like excessive use of nitrogen-based fertilizers, burning of crop residues, and livestock production generate significant greenhouse gas emissions, including methane and nitrous oxide. These emissions intensify global warming, which in turn affects agricultural productivity through erratic rainfall patterns, rising temperatures, and increased frequency of extreme weather events. Thus, agriculture is both affected by and contributes to environmental degradation, creating a cycle of vulnerability. However, alongside these challenges, agriculture also offers significant potential solutions for promoting environmental sustainability. The adoption of sustainable farming practices can substantially reduce the ecological footprint of agricultural production. Techniques such as organic farming, crop diversification, crop rotation, and integrated pest management help maintain soil health, reduce chemical dependence, and conserve biodiversity. Similarly, improvements in irrigation efficiency through drip and sprinkler systems can reduce water consumption while maintaining or even increasing productivity. Agriculture can also contribute positively to climate change mitigation and adaptation. Practices such as conservation agriculture, agroforestry, and sustainable land management enhance carbon sequestration, improve water retention, and increase resilience to climate shocks. The use of renewable energy sources such as solar-powered irrigation and bioenergy further reduces dependence on fossil fuels. By integrating environmental considerations into agricultural development, it is possible to balance productivity growth with ecological preservation. The dual role of agriculture—as both a contributor to environmental problems and a key part of their solution—highlights the importance of rethinking development strategies. Sustainable agricultural practices not only help protect natural resources but also support long-term economic growth by ensuring food security, stable livelihoods, and resilience to environmental risks. This integrated approach aligns economic objectives with environmental sustainability and social well-being. In conclusion, sustainability and efficient resource use are indispensable components of modern development strategies. Agriculture, given its close interaction with natural resources, occupies a strategic position in achieving sustainable development. Recognizing and strengthening its potential to act as an environmental solution rather than merely a source of degradation is essential for designing a balanced and inclusive growth model. Such an approach ensures that economic development remains viable, equitable, and environmentally sustainable in the long run.

Objectives

The specific objectives of this section/study are as follows:

- To examine the relationship between agriculture and natural resource use, particularly land, water, and environmental quality.
- To analyse the environmental challenges associated with agricultural practices, including resource depletion, environmental degradation, and greenhouse gas emissions.
- To assess the impact of unsustainable agricultural practices such as excessive water use, inefficient irrigation, and overuse of chemical inputs.
- To explore the potential of sustainable agricultural practices in reducing environmental stress and promoting efficient resource management.
- To evaluate the dual role of agriculture as both a contributor to environmental challenges and a key solution for achieving sustainability.
- To highlight the importance of integrating sustainability into agricultural development strategies for long-term inclusive and sustainable economic growth.

Methodology

The present study adopts a descriptive and analytical research design to examine the changing nature of economic growth and assess the role of agriculture in promoting inclusive and sustainable development. The methodology is designed to integrate theoretical perspectives with empirical patterns in order to evaluate the need for a new growth model in the contemporary economic context.

A. Nature of Study

The study is analytical in nature, focusing on understanding sectoral growth dynamics, employment challenges, and sustainability concerns within the framework of economic development. It relies primarily on secondary data and existing literature to draw inferences about growth patterns and development strategies, particularly in the Indian context.

B. Sources of Data

The study is based entirely on secondary data, collected from a wide range of reliable and authoritative sources. These include: Published research papers, books, and academic journals related to economic development, agriculture, and sustainability Reports and publications of national and international institutions such as government agencies, planning bodies, and development organizations Statistical data from sources such as economic surveys, census reports, and sectoral databases Policy documents and working papers related to agriculture, employment, and environmental sustainability These sources provide both qualitative and quantitative insights into long-term growth trends and sectoral performance.

C. Scope of Study

The study focuses primarily on India, while also drawing relevant comparisons from global development experiences where necessary. Special attention is given to: Sectoral composition of growth (agriculture, industry, and services) Patterns of structural transformation Employment generation across sectors Sustainability and resource use in agriculture The time scope of the study spans multiple phases of India's economic development to capture long-term structural changes.

D. Analytical Framework

The analysis is carried out using a sectoral and thematic approach. The study examines: Traditional and modern theories of economic growth The evolving role of agriculture in development Employment outcomes associated with different growth patterns Environmental implications of existing growth models Comparative analysis is employed to study variations in growth paths across different Indian states, emphasizing the sequencing of agricultural, industrial, and service-led development.

E. Tools and Techniques of Analysis

The study utilizes qualitative analytical tools, including: Conceptual analysis of development theories Comparative analysis of sectoral growth patterns Trend analysis based on available secondary data Logical reasoning and interpretation to synthesize findings Tables, conceptual diagrams, and descriptive summaries are used to present patterns clearly and support analytical arguments.

F. Limitations of the Methodology

As the study relies on secondary data, it is subject to limitations related to data availability, data consistency, and time lag in published sources. Moreover, the analysis is largely macro-level and does not involve primary surveys or econometric modeling, which may limit the scope for micro-level or causal inference.

G. Justification of the Methodology

The chosen methodology is appropriate for the objectives of the study as it allows a comprehensive examination of long-term growth trends, sectoral linkages, and sustainability issues. By integrating theoretical insights with empirical patterns, the methodology enables a holistic understanding of the changing role of agriculture and the need for a new development paradigm.

Results and Findings

1. Agriculture as a Growth Driver The study finds that agriculture continues to play a crucial role in economic growth, particularly during the early and intermediate stages of development. Agricultural growth has a strong impact on poverty reduction due to its ability to generate direct and indirect employment, especially in rural areas where a large proportion of the population depends on agriculture for livelihood. Improvements in agricultural productivity raise farm incomes, enhance food security, and increase purchasing power in rural regions. The findings indicate that agriculture-led growth creates strong forward and backward linkages with industry and services. Demand for inputs such as fertilizers, machinery, transport, storage, and processing expands industrial activity, while higher rural incomes stimulate demand for services such as education, healthcare, and trade. As a result, agriculture acts not only as a production sector but also as a catalyst for overall economic development. Countries and regions that prioritize agricultural development tend to achieve more inclusive

growth outcomes.

2. Sectoral Imbalance in Modern Growth One of the significant findings of the study is the presence of growing sectoral imbalance in modern growth patterns. While industrial and service sectors have contributed substantially to output growth, their capacity to generate employment has declined. The increasing use of automation and capital-intensive technologies has widened the gap between output growth and employment generation, particularly in industry and modern services. The study finds that labour has moved away from agriculture faster than non-agricultural sectors have been able to absorb it. This has resulted in underemployment, informal employment, and rising urban stress. The imbalance reflects an incomplete and uneven process of structural transformation, where growth occurs without adequate job creation. The inability of industry and services to absorb surplus labour reinforces the importance of agriculture as a stabilizing sector for employment.

3. Income Inequality and Sectoral Disparities The findings reveal that differences in productivity levels across agriculture, industry, and services have contributed significantly to income inequality. Agricultural workers often earn lower incomes compared to workers in industrial and service sectors due to lower productivity and limited access to technology, finance, and markets. This productivity gap has led to widening income disparities between rural and urban areas as well as across regions. Service-led growth, though contributing to higher average incomes, has benefited a relatively smaller and more skilled segment of the population. In contrast, agriculture supports a much larger workforce but offers comparatively lower returns. States and regions that bypass agricultural development and directly move toward service- or industry-led growth experience higher income inequality and regional imbalances. The findings suggest that improving agricultural productivity is essential for narrowing income disparities and achieving inclusive growth.

4. Sustainability and Resource Use Concerns The study identifies unsustainable use of natural resources as a major challenge to long-term economic growth. Excessive extraction of groundwater, inefficient irrigation practices, and overuse of chemical fertilizers have resulted in land degradation, declining soil fertility, and water scarcity. These environmental stresses threaten the long-term viability of agriculture and undermine food security. At the same time, the findings highlight agriculture's potential role in promoting sustainability. Adoption of sustainable farming practices, efficient water management systems, and renewable energy use can significantly reduce environmental degradation. Agriculture thus plays a dual role—as a contributor to environmental challenges and as a key part of the solution. Sustainable agricultural development is therefore essential for ensuring that economic growth remains environmentally viable over the long run.

Discussion

The findings of this study provide important insights into the changing nature of economic growth and support the argument that traditional development models require re-evaluation in the contemporary context. The results highlight the continued relevance of agriculture in economic development, particularly in relation to employment generation, income distribution, and sustainability. These findings align with modern development literature, which emphasizes inclusive and balanced growth rather than growth driven solely by output expansion.

The finding that agriculture acts as a significant growth driver, especially in the early stages of development, reinforces the arguments made by several development economists who challenge the notion of agriculture as a passive or declining sector.

Agricultural growth has a strong poverty-reducing effect because it directly affects rural livelihoods, where a large proportion of the poor reside. The strong forward and backward linkages identified in this study confirm that agriculture stimulates industrial and service-sector activities through increased demand for inputs, processing, transportation, and consumption. This suggests that neglecting agriculture can weaken the overall growth process and limit its inclusiveness.

The presence of sectoral imbalance observed in the findings reflects a major limitation of modern growth patterns. While

industry and services have emerged as key contributors to GDP, their limited capacity to generate employment has intensified the problem of jobless growth. This finding supports the argument that reliance on capital-intensive industrialization and skill-intensive services alone cannot address the employment needs of developing economies. The inability of these sectors to absorb surplus labour released from agriculture explains the persistence of informal employment and underemployment. In this context, agriculture plays a crucial stabilizing role by continuing to provide livelihoods, thereby reducing social and economic distress.

The study further reveals that income inequality is closely linked to sectoral productivity differences. Higher productivity and wages in industry and services compared to agriculture have widened income disparities between rural and urban areas and across regions.

Service-led growth, although contributing to higher average incomes, tends to benefit a smaller and more skilled segment of the population. This finding supports earlier research suggesting that growth paths which bypass agriculture or prioritize services prematurely often result in unequal development. Enhancing agricultural productivity and integrating it with industrial growth can help reduce these disparities by raising rural incomes and distributing growth benefits more evenly.

Another critical aspect discussed in the findings is sustainability and resource use, which poses a major challenge to long-term growth. The study confirms that unsustainable agricultural practices have contributed to environmental degradation through excessive water use, chemical inputs, and greenhouse gas emissions. However, the discussion also highlights agriculture's dual role by recognizing its potential to contribute to environmental solutions. Sustainable farming practices, efficient irrigation technologies, and renewable resource use can not only reduce environmental stress but also improve resilience and long-term productivity. This dual role underscores the importance of integrating environmental considerations into development strategies rather than treating sustainability as a secondary concern.

Overall, the discussion suggests that the sequence of development matters significantly. Growth strategies that prioritize agriculture, followed by gradual industrial expansion and supported by services, appear to generate more inclusive and sustainable outcomes. The Indian experience, as reflected in the findings, demonstrates that premature shifts toward industry or services without a strong agricultural base often result in employment constraints, inequality, and regional imbalances. Thus, the study supports the case for a new growth paradigm that recognizes agriculture as a central component of economic transformation.

In conclusion, the discussion reinforces the core argument of the study—that inclusive and sustainable development requires a departure from conventional growth models. Integrating agriculture into the broader growth framework, addressing employment challenges, and promoting sustainable resource use are essential for ensuring that economic growth translates into widespread social and economic well-being. The findings and discussion together underline the need for policy strategies that emphasize balanced sectoral development rather than relying on narrow, output-focused growth approaches.

Recommendations

Based on the findings and discussion of the study, the following policy-oriented and strategic recommendations are proposed to promote inclusive and sustainable economic development:

1. Promote Agriculture-Led Development in the Early Stages The study strongly recommends prioritizing agriculture-led growth, particularly in the early stages of development. Strengthening agriculture can generate broad-based employment, reduce poverty, and create demand for industrial and service-sector activities. Public investment in agricultural productivity, extension services, and farmer support mechanisms should be enhanced to ensure stable rural incomes. A strong agricultural foundation can serve as a catalyst for balanced structural transformation and inclusive growth.

2. Encourage Labour-Intensive Industrial Development To address the challenge of jobless growth, there is a need to promote labour-intensive industries such as agro-processing, textiles, food processing, and small-scale manufacturing. Policies should encourage industries that can absorb surplus labour released from

agriculture rather than relying solely on capital-intensive production methods. This would help bridge the gap between economic growth and employment generation and reduce dependence on informal employment. 3. Invest in Sustainable Farming Practices Given the environmental constraints facing modern growth, investment in sustainable agricultural practices is essential. Adoption of eco-friendly technologies such as water-efficient irrigation, organic farming, crop diversification, and integrated pest management should be encouraged through incentives and awareness programs. Sustainable farming not only conserves natural resources but also enhances long-term productivity and resilience against climate change. 4. Improve Rural Infrastructure and Market Access Strengthening rural infrastructure is critical for linking agriculture with industry and services. Investments in roads, storage facilities, cold chains, transport, and digital connectivity can reduce post-harvest losses and improve farmers' access to markets. Better market integration ensures fair prices for agricultural produce, increases rural incomes, and promotes value-added activities such as processing and packaging. 5. Support Technological Innovation in Agriculture Technological innovation should play a central role in transforming agriculture into a modern, dynamic sector. Promoting research and development, mechanization suitable for small farmers, use of digital platforms, and climate-resilient technologies can significantly enhance productivity and efficiency. Technology adoption must be inclusive and accessible to small and marginal farmers to prevent widening inequalities within the agricultural sector. 6. Integrate Agriculture with Industrial and Service-Sector Policies Development strategies should adopt an integrated approach that recognizes the interdependence between agriculture, industry, and services. Agro-based industries, rural non-farm employment, and service activities linked to agriculture should be actively promoted to ensure smooth structural transformation and balanced growth.

Conclusion

This study concludes that the conventional models of economic growth are increasingly inadequate in addressing the contemporary challenges of employment generation, income inequality, and environmental sustainability. Traditional development strategies, which emphasize a linear shift from agriculture to industry and services, fail to fully capture the complexities of modern economic transformation. The evidence discussed in this study highlights the urgent need for a new growth model that places inclusivity and sustainability at its core, rather than focusing solely on output expansion. The analysis clearly demonstrates that agriculture continues to play a vital role in economic development, particularly in developing economies such as India. Far from being a stagnant or backward sector, agriculture has the potential to act as a powerful engine of growth by generating employment, reducing poverty, stimulating demand, and creating strong linkages with industry and services. The findings confirm that agricultural growth has a more direct and widespread impact on livelihoods, especially in rural areas, making it essential for achieving inclusive development. The study also reveals that growth driven predominantly by industry or services has led to sectoral imbalances, where output growth has not translated into corresponding employment generation. Capital-intensive industrialization and skill-intensive service-sector expansion have contributed to jobless growth and rising income inequality. These outcomes underline the limitations of growth strategies that bypass agriculture or relegate it to a secondary role. Furthermore, the study emphasizes that sustainability and efficient use of natural resources are indispensable components of long-term development. While agriculture has contributed to environmental challenges through unsustainable practices, it also offers significant solutions through sustainable farming methods, efficient water management, and climate-resilient technologies. This dual role reinforces the need to integrate environmental considerations into economic planning rather than treating sustainability as a separate or secondary concern. Overall, the study supports a sequential development approach, beginning with agriculture-led growth, followed by industrialization and supported

by the expansion of services. Such a strategy ensures balanced structural transformation, stronger employment outcomes, reduced inequality, and sustainable use of resources. The Indian experience, as well as broader development evidence, suggests that this sequence provides the most effective and resilient pathway to long-term economic development. In conclusion, the study advocates for a reoriented growth paradigm that recognizes agriculture as a central pillar of development, supported by industry and services in an integrated manner. Policymakers must move beyond narrow sector-specific approaches and adopt holistic strategies that align economic growth with social equity and environmental sustainability. Only through such an inclusive and balanced framework can sustainable economic development be achieved in the long run.

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